

Energy from biomass

We are going to learn about biomass energy which is another important renewable energy resource.

Let's begin with what is biomass?

Biomass is natural and organic materials or wastes produced by plants or animals. Mostly biomass is derived from plants or plant-based materials which are referred as lignocellulosic biomass. Even animal-derived materials are used for biomass generation.

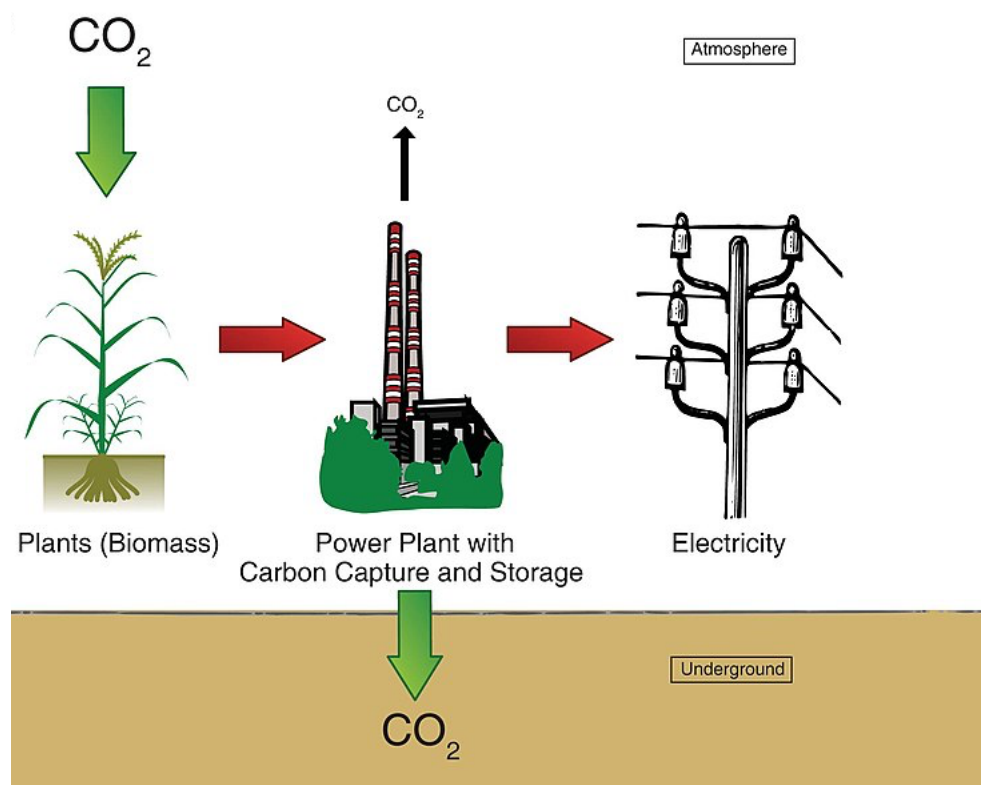


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Biomass can be used to generate three important products

First one is Biopower or bioenergy: Biomass can be converted into heat and electricity using various techniques.

Second one is Biofuels Biomass can be converted into biofuels such as bioethanol and biodiesel

Third one is biogas decomposition of biomass will give biogas which can be used to generate either heat or electricity.

Let's discuss above mentioned three important products one by one.

First biopower or bioenergy

How to harvest bioenergy such as heat or electricity from biomass?

Harvesting energy from biomass is not a new concept.

Biomass has been in use since the stone age and cavemen used wood material to burn to cook his food and keeping warm.

Strictly speaking biomass energy is indirect solar energy. Solar energy is trapped by green plants through photosynthesis and converted into biomass energy.

As I mentioned earlier biomass is composed of plant-derived materials such as starch and sugar-producing plants wood crop residues latex producing plants vegetable oilseed plants agricultural wastes forestry waste and various biomass wastes such as municipal wastes industrial solid wastes sewage waste agricultural wastes industrial effluents and cattle dung apart from this biomass residues such as wheat straw maize cubes rice straw groundnut wastes jute sticks and so on are source for energy.

Biomass can be converted into either heat or electricity by various processes. They are combustion digestion pyrolysis fermentation and catalyzed reactions.

Let's discuss first process called **biomass combustion** means directly burning organic materials. For ages humans have utilized this technology to create fire and the heat generated can be efficiently used to generate electricity through steam. The plant or animal-derived natural products including wood crop residues cattle dung sewage wastes and agricultural wastes are the most commonly used materials which can be burned effectively to generate heat and this heat energy can convert water into steam. This steam hits the turbine blades causing turbine blades to rotate. The rotation of the turbine in turn energies a generator makes electricity.

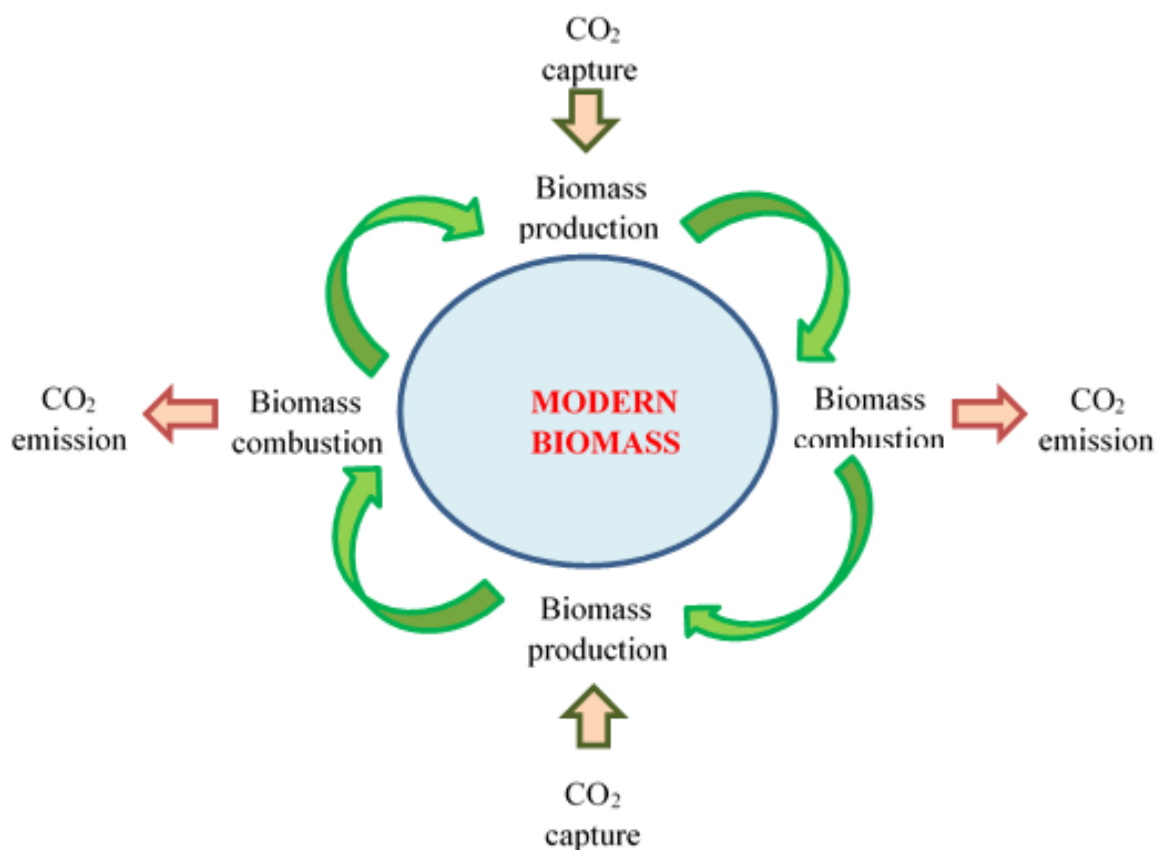


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Second process is **Digestion**. Digestion is another technique that makes use of existing biodegradable waste materials. Most of the time the digestion process will be carried out in the absence of oxygen. That is known as anaerobic digestion. During the digestion process biomass will be broken down into gases like methane carbon monoxide etc. with the help of

microorganisms such as bacteria. Actually, microorganisms feed the biowastes and decompose them. During the process the above-mentioned gases will be released.

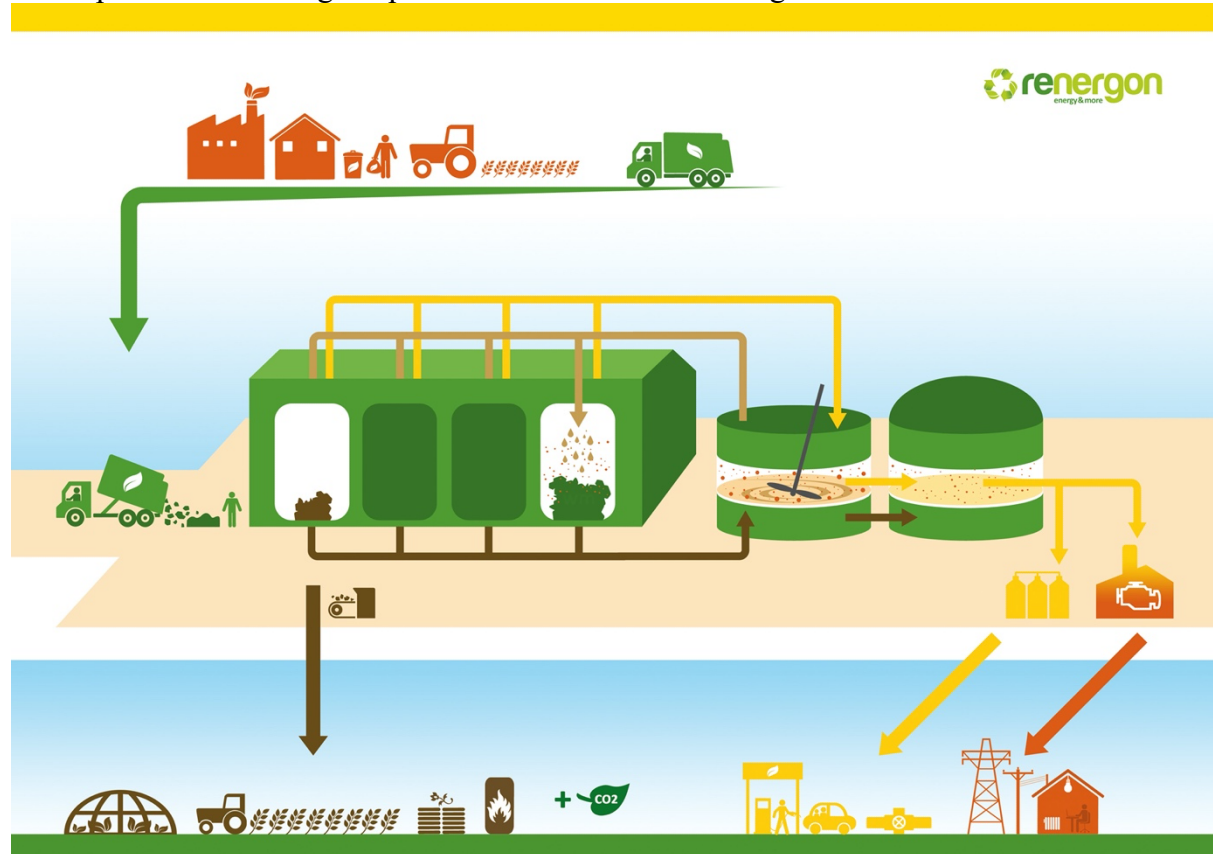


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The third process is **pyrolysis** which means the thermochemical decomposition of natural biomass at elevated temperatures in the absence of oxygen or water. During the process biomass can be converted into mostly solid materials like charcoal. This process is irreversible. At the industry level these pyrolysis processes will be done under pressure and at temperatures greater than 800° F. Industrial pyrolysis can produce even liquid fuel. End product of the pyrolysis usually charcoal which have double the energy as that of original biomass and can be transportable and more efficient.

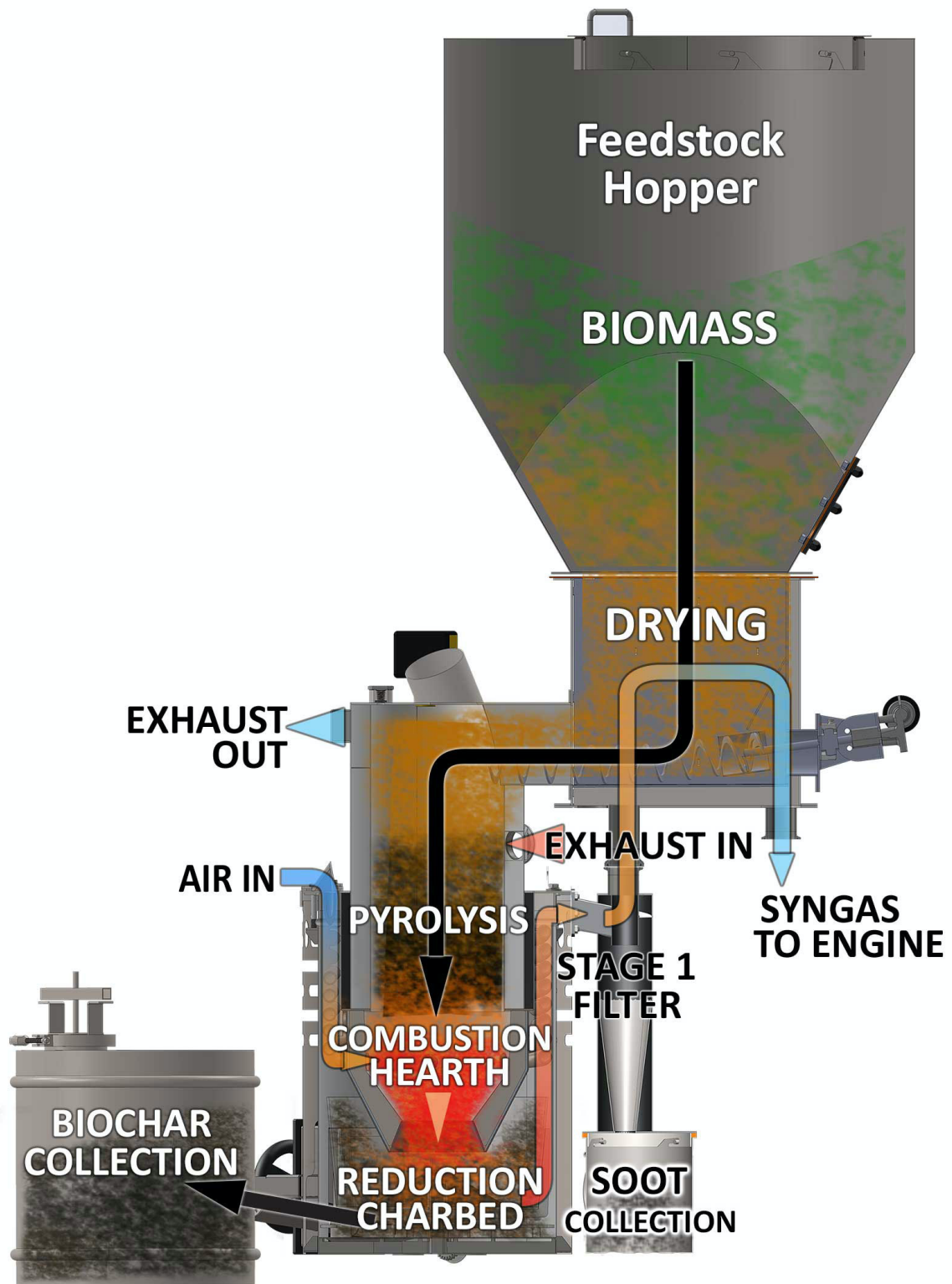
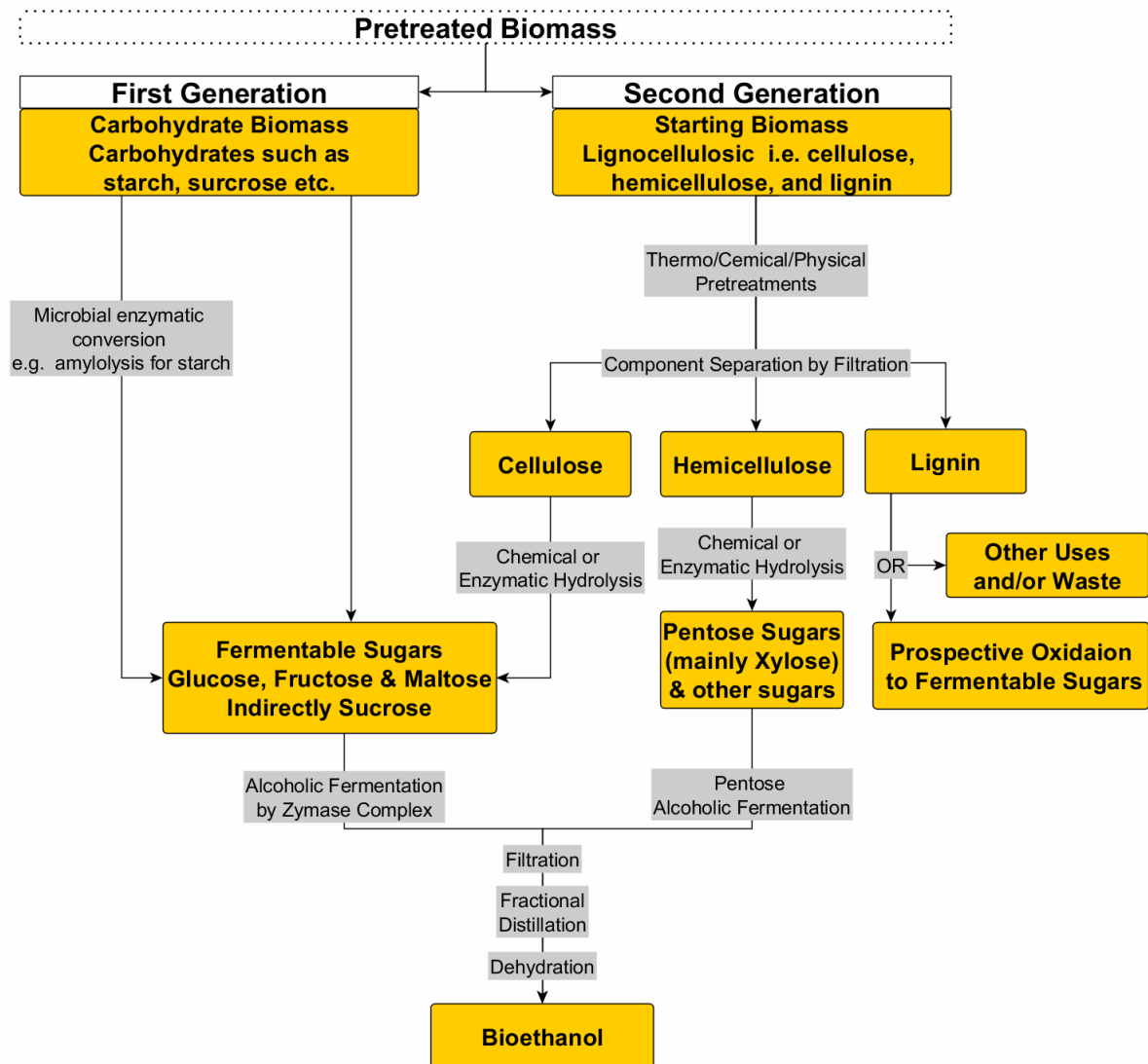


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Next process is **Fermentation**: Fermentation is a complex biochemical process that converts sugar into alcohol or acids or other gases. The fermentation process occurs mostly in the presence of yeast and bacteria.

Mostly the thermal conversion or combustion process is the dominant mechanism to convert biomass into heat or electricity. The energy produced by burning biomass is predominantly

well suited for countries where the fuelwood grows more rapidly for example tropical countries.



Let's discuss another important biomass energy source that is **bioethanol**

Bioethanol can be produced by converting biomass into liquid fuels to meet transportation needs. Bioethanol can be created by using fermentation of organic materials.

Microorganisms such as bacteria and yeasts play an important role in fermenting starchy sugary biomass products like sugarcane corn sorghum and yielding products like biofuels and bioethanol.

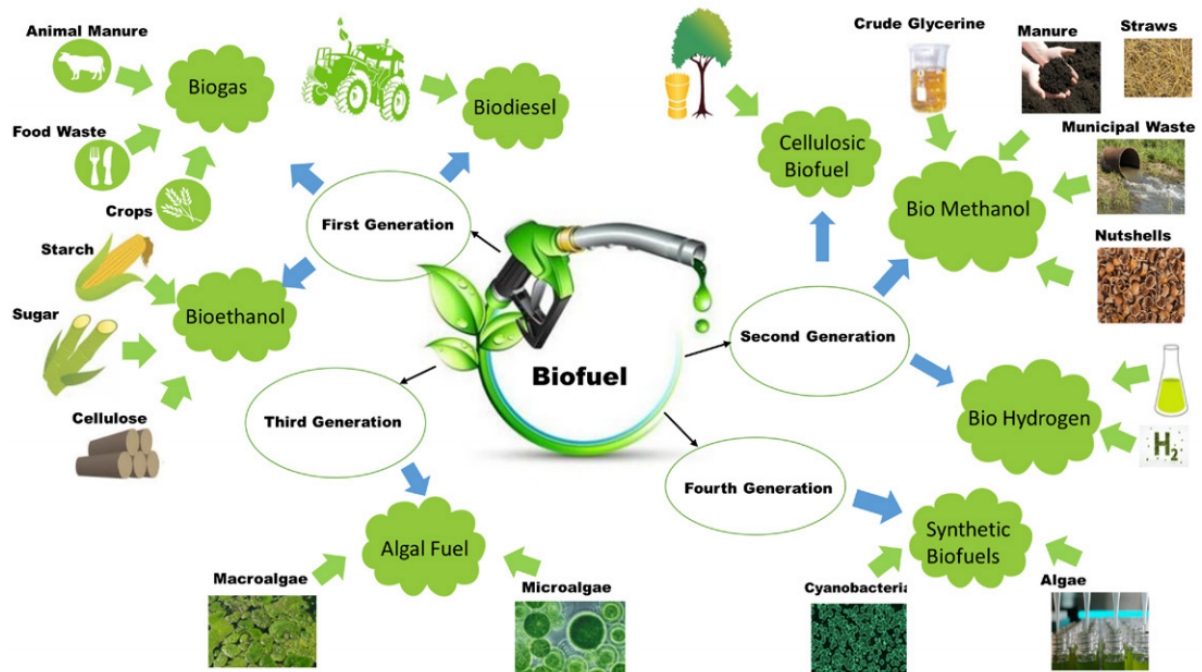


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Bioethanol can be used as fuel in a variety of applications.

Bioethanol cannot be used alone in fuel cars. Instead, bioethanol can be blended with gasoline. The most widely used blends are E10 and E85. The E and number refer to the percentage of ethanol fuel by volume in the blend. For examples E10 is a blend of 10% ethanol and 90% gasoline. In the case of E85 85% ethanol and 15% gasoline.

Bioethanol is a clean fuel; even though its calorific value is lesser than gasoline bioethanol increases the octane number and decreases the amount of greenhouse gas emissions. Bioethanol production will offer economic benefits to farmers. Most of bioethanol used today comes from sugarcane and corn. Many bioethanol production plants are farmer-owned plants. It is indirectly supporting the industry's growth. Bioethanol actually reduces the fuel economy by 2 to 3 percent. Hence it is a desirable alternative fuel leading the future away from our gasoline addiction.

Common ethanol fuel mixtures

Code	E5	E10	E15	E25	E85	E100
Composition	max 5% anhydrous ethanol min 95% gasoline	max 10% anhydrous ethanol min 90% gasoline	max 15% anhydrous ethanol min 85% gasoline	max 25% anhydrous ethanol min 75% gasoline	max 85% anhydrous ethanol min 15% gasoline	~5.3% water 100% Brazilian hydrous ethanol (contains on average 5.3 vol.% water)
Countries	Western Europe today	USA today (Western Europe in near future)	USA EPA approval cars > 2000	Brazil	USA / Europe	Brazil

Gasoline blends for use in regular cars

Flex Fuel Vehicles

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Next, we can discuss about **Biodiesel**.

Biodiesel is an alternative liquid fuel especially in the transportation sector. Biodiesel is a potentially low-carbon-containing renewable biodegradable fuel sharing similar chemical properties and physical properties to conventional petroleum diesel. Chemically these are long-chain fatty acid esters produced from sources like vegetable oil and animal fats and from an alcohol such as methanol or ethanol produced from plants and plant wastes using a catalyst. It shows great promise as fuel when used alone or when mixed with petroleum diesel.

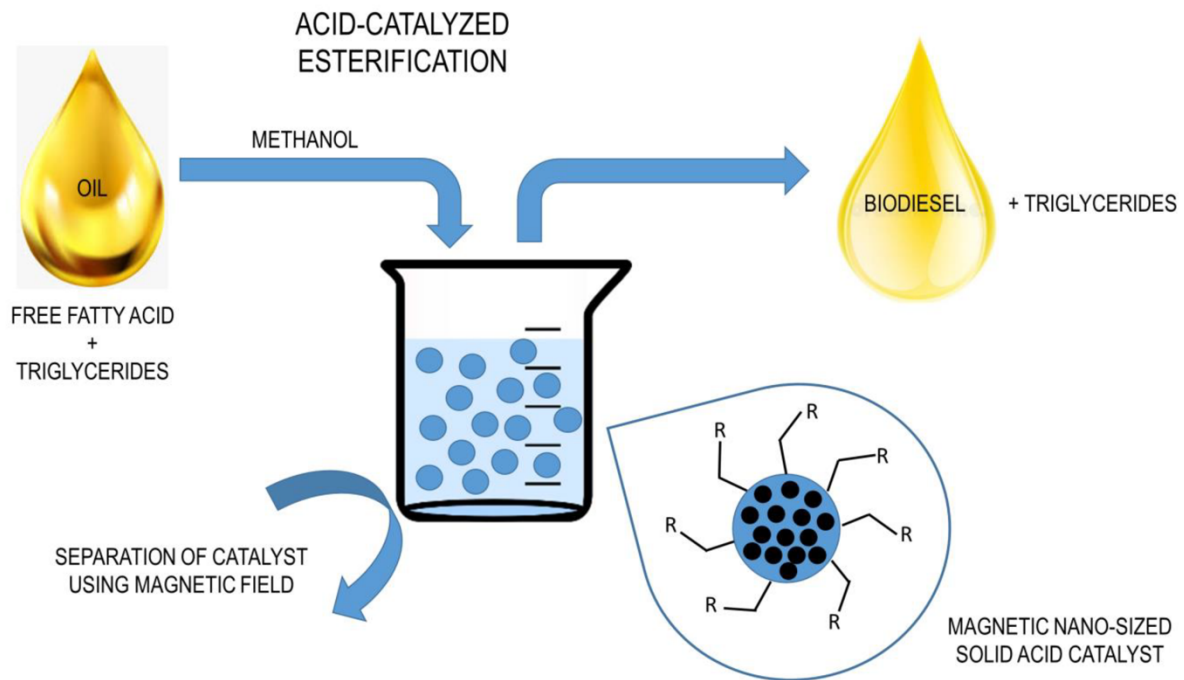


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Most of the time these vegetable oils are extracted from rapeseeds soybeans oil palms sunflowers jatropa shrubs and animal fats. In India biodiesel is mainly produced from the jatropa shrub a plant growing mostly in tropical countries like India Brazil and Africa. The important advantage is Jatropa crops grow even in hot dry tropical areas. They don't need frequent rain and fertilizers.

Indian Railway has decided to use biodiesel in a big way in locomotives to reduce fuel consumption and cut down its oil bill.

Next, we can discuss about **Biogas**

Biogas is a mixture of gases like methane carbon dioxide hydrogen sulfide and moisture produced by the breakdown of biodegradable wastes such as agricultural wastes plant residues municipal wastes food wastes.

Biogas is a mixture of gases produced from the decomposition of organic matter in the absence of oxygen. The main constituent of biogas is methane nearly 65% and the rest is a mixture of carbon dioxide hydrogen and hydrogen sulfide. It is produced by anaerobic degradation of various wastes due to microbes and biochemical reactions. That's why a biogas plant is often known as an anaerobic digester.

Most of the time the biogas plant comprises a dome-like structure. In this energy crops such as corn or maize silage or biodegradable wastes including sewage wastes and discarded food residues fats sludge cow dung etc. along with wastewater are fed into the digester through the inlet. The digester is basically an air-tight tank where the anaerobic decomposition of organic matter occurs. During the process the microorganisms transform biomass waste into biogas primarily methane and carbon dioxide and digestate. After that we can draw these gases through pipes from the storage tank above the digester and This biogas can be used for

heating producing electricity and for many other applications. The digestate which is the left out organic matter can be used as a bio-fertilizer and spread directly onto fields.

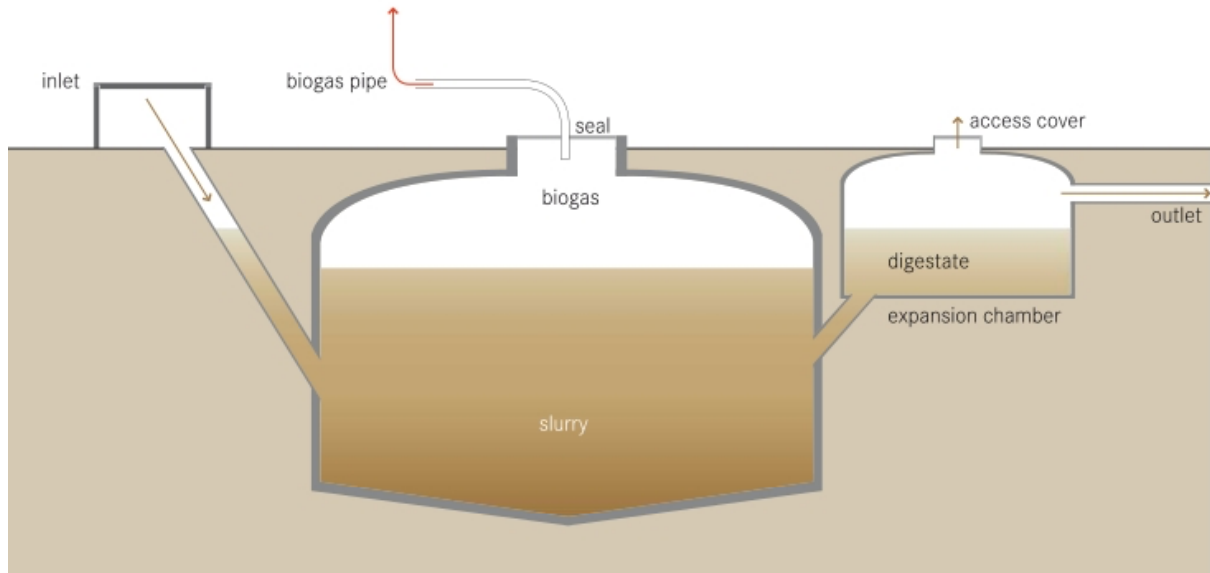


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Biogas has a lot of advantages.

It is a non-polluting clean and low-cost fuel. A Biogas plant does not require an expensive installation cost.

Very much useful in rural areas since a lot of animal waste is available

India is the top country by cattle population in the world

There is no storage problem since biogas is directly supplied from the plant

The sludge or digestate leftover is rich in fertilizer

Biogas burns without smoke; therefore, it evolves no harmful gases such as CO₂ CO NO₂ and SO₂.

Biogas production depends on the amount of waste produced. Of course, waste production is an endless process in this current population growth.

There are a few disadvantages of biogas. Biogas production is inefficient on a large scale, and it is not economically feasible to use on a large scale. If a large amount of methane leaks it will cause a greenhouse effect and hence global warming.

Let's discuss about the difference between biomass and other fossil fuels

Biomass energy or biofuels can be generated quickly but fossil fuel formation takes millions of years hence it is renewable quickly

Biomass takes carbon out of the atmosphere; hence net CO₂ emissions are low compared with fossil fuels.

Finally, now we can discuss the advantages and disadvantages of biomass energy.

There are plenty of benefits due to biomass energy

Biomass is always and widely available because the production of biomass is infinite; since we consistently produce wastes such as garbage sewage waste etc. biomass energy is considered as a renewable source of energy.

Biomass energy production maintains carbon neutrality. During the photosynthesis process carbon dioxides are absorbed from the atmosphere by plants. ; When we burn biomass fuels the same amount of carbon dioxides will be released and hence it helps in maintaining carbon neutrality.

Fossil fuel production requires a lot of investment in oil drilling extractions and installing gas pipelines and distillation, but biomass energy production is much cheaper.

Farmers and local investors can generate higher profits from a lower output.

Biomass production adds a revenue source for farmers.

By burning solid wastes, we can reduce the landfills requirement by 60 to 90 percent. and thus it reduces the cost of waste disposal methods like sanitary landfills.

The use of biomass energy greatly reduces greenhouse gas emissions.

The use of biomass can reduce our dependency on fossil fuels because biofuels are the best alternative renewable liquid fuels available for the transportation sector.

There are also some shortcomings including

Some biofuels like bioethanol and biodiesel are relatively inefficient as compared to fossil fuels.

Biomass energy production may increase the risk of deforestation in the future.

It is difficult to find a convenient place for biomass energy plants near the urban area because people are afraid of the toxic and harmful gas emissions.